

KALPITIYA-KARAITIVE-PUTALLAM (KKP) & SERUVILA-ALLIA-ULLAKALIE (SAU) COASTAL WETLAND COMPLEXES

A COMPREHENSIVE GIS BASED ASSESSMENT AND SUGGESTIONS FOR CONSERVATION ACTION



-14th Annual Scientific Sessions of the Sri Lanka Association for Fisheries and Aquatic Resources -

Presented by: Chandee Corea

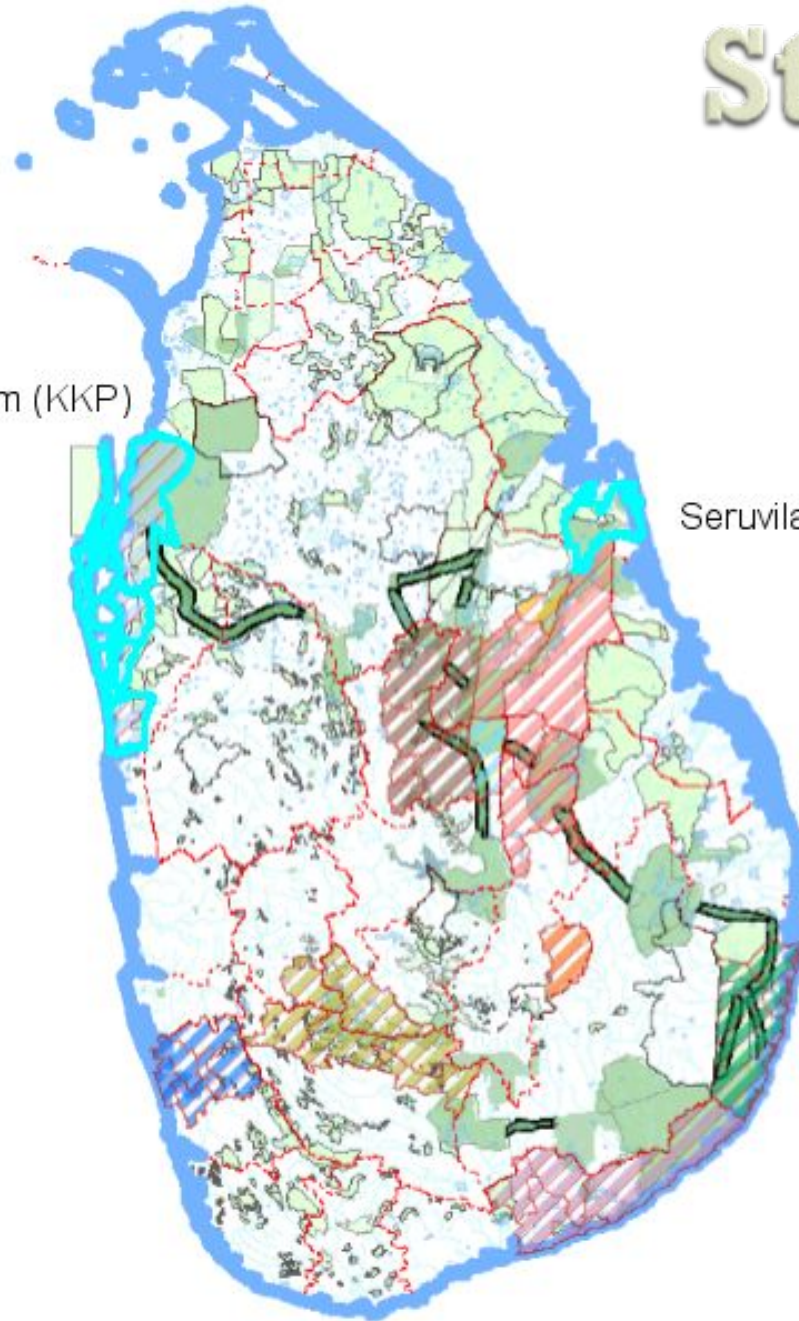
Summary

- The study of water in landscapes is increasingly becoming essential to understand the complexities of water as it flows from watershed to ocean.
- Integrating the unifying themes of Hydrology and Ecology to see the connectivity between landscapes and riverscapes is key to understanding human impacts.
- Knowledge of these issues allows us to effectively address poverty, food security, health hazards , natural disasters , water security and environmental concerns.

Study Sites

Kalpitiya-Karaitive-Puttalam (KKP)

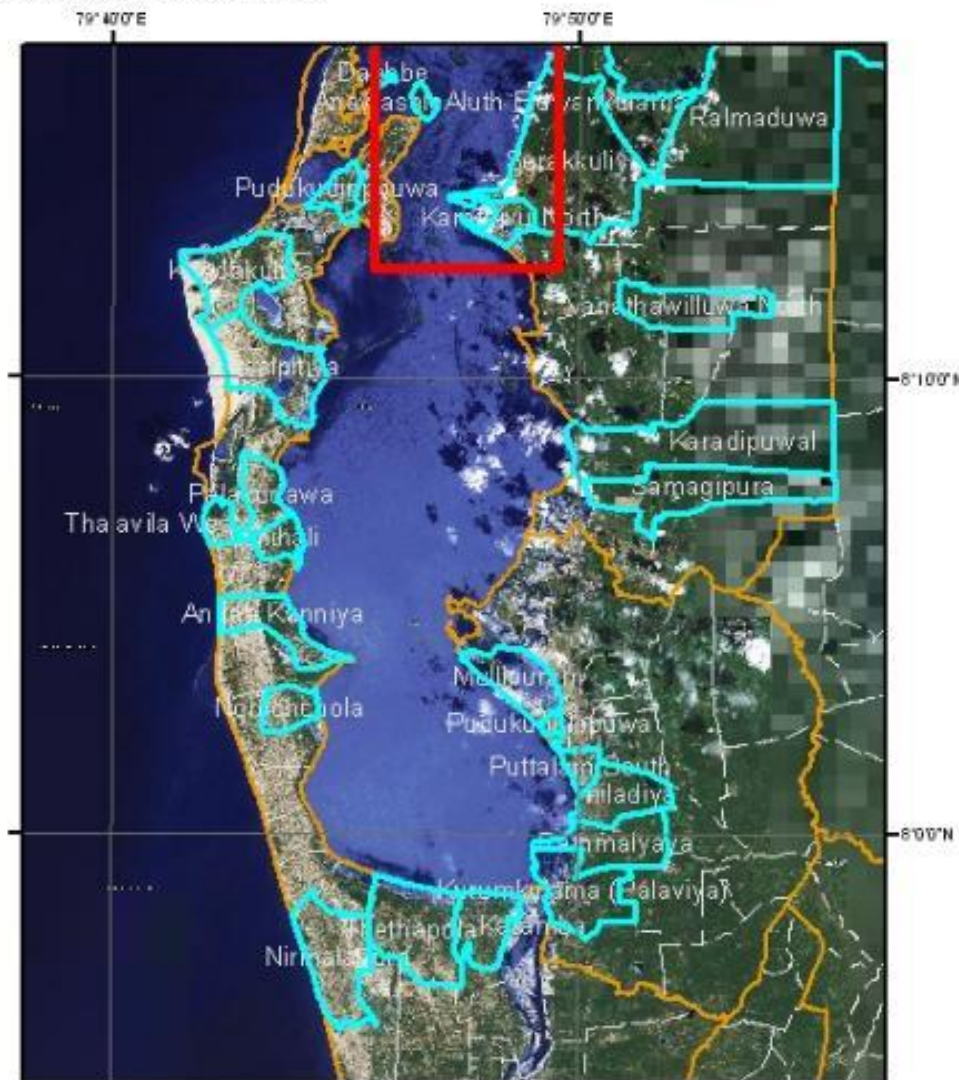
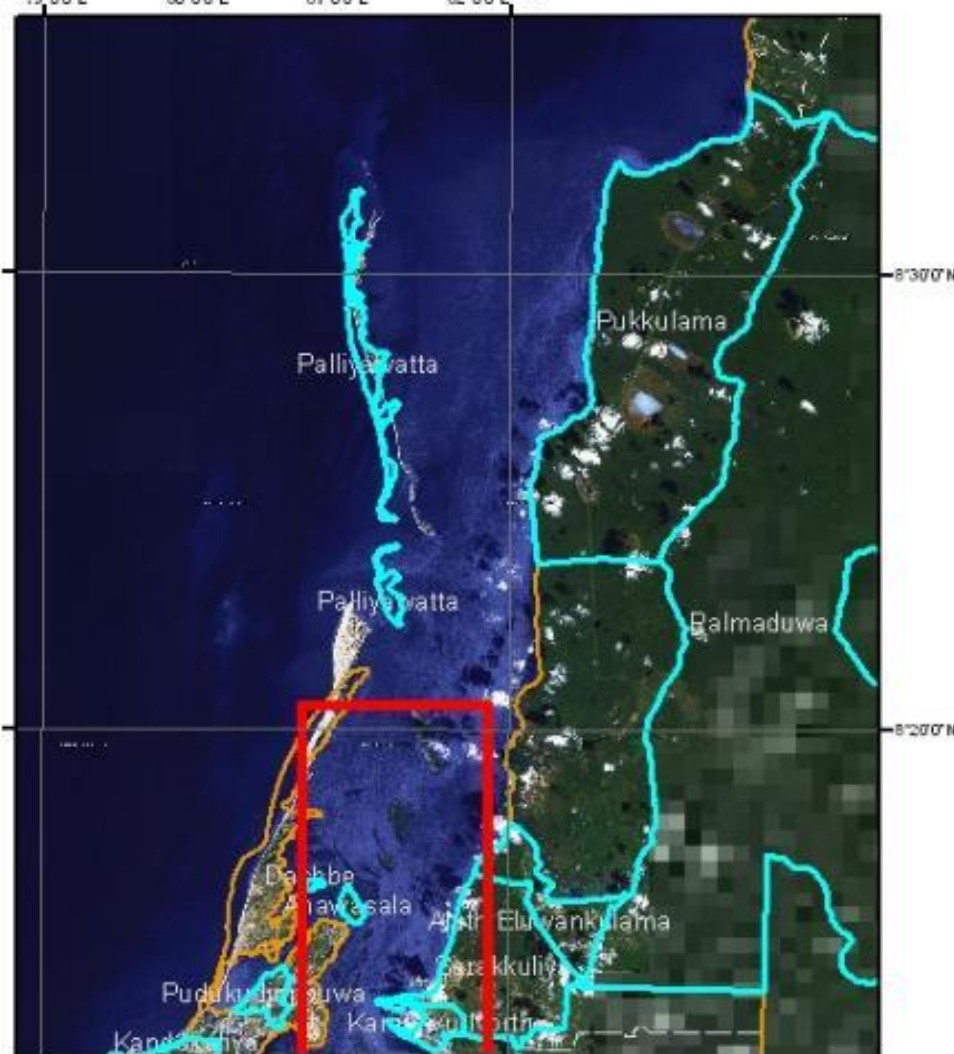
Seruvila-Allai-Ullakallie (SAU)





Kalpitiya-Karaitive-Puttalam Coastal Wetland Complex

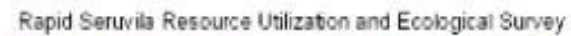
Villages (GND's) Surveyed in
3 Divisions (DSD's) of the North-Western Province



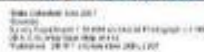
SLWCS Data: Collected: Dec 2005 to Oct 2006
Satellite Imagery:
1999-01-01, "MODIS EarthSat, 1,100,000, 15.0m, "Color"
Boundaries: Urban Development Authority,
Survey Department and LacNET, Sri Lanka
Published on: September 8th, 2007

0 1 2 4 6 8 Kilometers
1:300,000

(C)(R) www.SLWCS.org GIS Lab
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doi:10.1016/j.sbsbs.2007.08.001



Methodology and Materials Used

- Sampling sites were selected randomly considering the representation of habitat types and accessibility.
- The *Broad Reconnaissance* Survey and *Point-Count* methods were used at both sites
- At KKP random line transects were also carried out.

	KKP	SAU
Number of field days	43	4
Survey Period	Dec 2005-Oct 2006	Dec 17 th -19 th 2007

Number of species observed		
	KKP	SAU
Butterflies	49	57
Dragonflies	8	Not surveyed
Amphibians	6	11
Reptiles	26	4
Mammals	this value must be incorrect because there are only 100 species mammals found in Sri Lanka including wales also . In slide no 11 you say 28 species mammals.	10
Birds	185	123
Plants	181	Not surveyed

Ecological Findings @ KKP

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- Important Records

- Threatened endemic Sri Lanka Wood Frog (*Rana gracilis*)
- Mourning Gecko (*Lepidodactylus lugubris*) which was recorded for the first time in the Puttalam area
- Large Salmon Arab (*Colotis fausta*) and the Crimson Tip (*Colotis danae*)



Ecological Findings @ KKP

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- No Live Turtles
- The shells of Green Turtle (*Chelonia mydas*) and Olive Ridley Turtle (*Lepidochelys olivacea*) were found frequently on the beaches.
- Most of these turtles had been killed for their meat.
- All communities were aware that it is illegal to kill turtles and there was a fine for hunting turtles and selling their meat and eggs



Figure : Shells of Poached Turtles; A Frequent Sight in the Islands



Ecological Findings @ KKP

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- **Four avian groups**

- Water birds, raptors, forest birds and waders

- **Several rare sightings**

- A flock of 17 Great Knots (*Calidris tenuirostris*)
- The Eurasian Oystercatcher (*Haematopus ostralegus*)
- First year juveniles of 12 bird species remained in the KKP throughout the year.



Dunlin (*Calidris alpine*).



Great Knot (*Calidris tenuirostris*).

Ecological Findings @ KKP

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- 28 species of mammals
- 18 families
- 1 species is endemic and 1 species is threatened.
- 5 domestic species that have become feral in the area.
- Jackal and Wild boar in offshore islands.



Figure : Puttalm Ass

Ecological Findings @ KKP

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- 5 marine mammal species including the dugong (*Dugong dugon*).
 - The dugong might become extinct in Sri Lanka due to the lack of committed conservation action.
 - The SLWCS is in the process of initiating a study of the dugong in Kalpitiya.
- The CEA (1994) recorded The Spinner Dolphin (*Stenella longirostris*), Pan tropical spotted Dolphin (*Stenella attenuata*), Bottle nosed Dolphin (*Tursiops truncatus*) and the Common Dolphin (*Delphinus delphis*)



Figure : Dugong (*Dugong dugon*).

Ecological Findings @ SAU

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- **Important Records**

- Endemic Toque Macaque
(*Macaca sinica sinica*)

- don't put Grey Langur as a
important sighting.

Ecological Findings @ SAU

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- **Important Records**

- Endemic Toque Macaque
(*Macaca sinica sinica*)

- Grey Langur
(*Semnopithecus priam*)

Methodology and Materials Used for Looking at Human Impact

- Rapid Rural Appraisal (RRA) and Participatory Rural Appraisal (PRA) tools were used.
- 40-50% of the GNDs in each DSD were selected randomly

KKP

3 months time period for field data collection, within that period 10 to 15 days each month were spent in the field to achieve 40% coverage at DSD level.

Total field days approx: 50

Number of Interviews (RRA Survey): 120

GNDs covered: 30

DSDs covered: 3

Community Leaders: 30

Community members: 90

Stakeholders (Govt. Departments and NGOs) met: 13

SAU

3 days time period for field data collection, within that period of three days each day was spent in the field to achieve 25% coverage at DSD level.

Total field days approx: 3

Number of Interviews (RRA Survey): 36

GNDs covered: 12

DSDs covered: 3

Community Leaders:

Community members:

Stakeholders (Govt. Departments and NGOs) met:

Human Impact

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

Coastal Habitat	Extent in the entire (ha)	Extent in the Puttalam District (ha)	Percentage of the total found in the Wetlands of the Puttalam District
Mangroves	8687	2264	26.06%
Salt marshes	23819	3461	14.53%
Sand dunes	7606	2698	35.47%
Estuaries	11800	2772	23.49%
Shore Line/Beaches	158017	39119	24.76%
Marsh	9754	2515	25.78%
Other	18839	3428	18.20%
Total	238522	56257	23.59%

Extent of coastal habitats (CCD
1997)

Human Impact

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- Many of the fishing methods practiced in the lagoon are destructive to the ecosystem.
- Fishing methods that either pull or drag nets along the lagoon floor are the most destructive.
- The ecological cost of push nets is approximately 12 times greater than its annual benefits.



Human Impact

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- Aquaculture –Reduction of Mangroves
 - The last two decades the shrimp culture industry rapidly expanded in the Puttalam District.
 - Much of this development took place along the West coast in the KKP/Mundel estuarine system
 - 63% (approximately 993 hectares) decline of the mangroves in the area between 1981 and 1992 (Senerath 1998).
 - The ecological and economical impact of the loss has not been studied or estimated
- Succession
 - structural modifications due to mangrove encroachment.
 - affects the extent of inter tidal mud flats
 - affects the animals that depends on it for their survival.



Mangrove Replantation –Turtle Conservation Project, Kandakuliya

Human Impact

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

By Catch

- The impacts are not known and have been hardly studied
- Just discarded on to the beach
- Can easily throw back into the seas
- The current understanding is that around 8% of the total global catch is discarded as by-catch.
- The accidental capture, or by catch, of dolphins, turtles and numerous other species is a serious issue
- The scale of by catch in fisheries can affect the structure and function of marine systems at the population, community and ecosystem levels.



BY-CATCH is the most serious environmental impact of modern commercial fisheries on marine

Human Impact

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

- Socio-Economic Status

- The 2001 census records 168,106 people in the 3 Divisional Secretariat Divisions.
- 10% of the population are war refugees
- 1.8% natural growth (IUCN, 2002).
- The Kalpitiya DSD had a migratory population of over 25,000 in 2001 (CRMP, 2006).
- Both the Kalpitiya and Puttalam DSDs are rated among the 10 poorest in Sri Lanka (DCS, 2002).

Human Impact

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

Number of DSDs	Total Number of GNDs	Sample GNDs	Sample GNDs (3 households/G.N.D)
1. Kalpitiya	31	14	<i>3 households (3*14) = 42</i>
2. Puttalam	22	8	<i>3 households (3*8) = 24</i>
3. Vanathavilluwa	17	8	<i>3 households (3*8) = 24</i>
Total	70	30	Total selected GNDs = 30 Total selected Households = 90 Total Interviews = 120

STAKEHOLDERS INTERVIEWED

The following stakeholder organizations were interviewed:

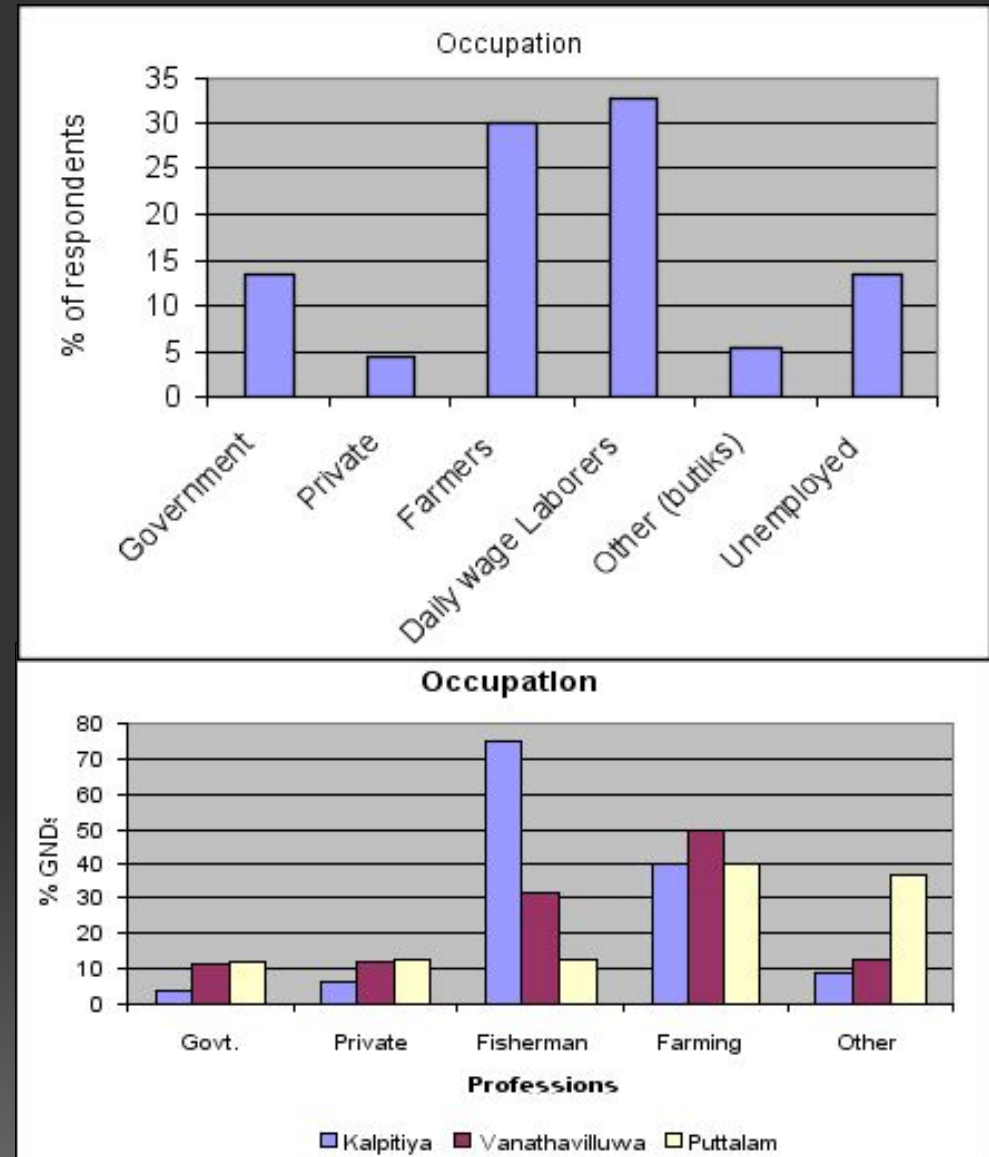
1. Central Environmental Authority (CEA/Government Organization)
2. Department of Wildlife Conservation
3. Divisional Secretariat Divisions (DSDs in charge)
4. Grama-Niladari Divisions (GND Officers)
5. International Union for Conservation of Nature (IUCN)
6. Turtle Conservation Project, Colombo, Kalpitiya
7. National Aquatic Resources Development Agency
8. Coastal Resources Management Project (Coastal Conservation Department)
9. Fisheries Department, Colombo

Human Impact @ KKP and SAU

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -

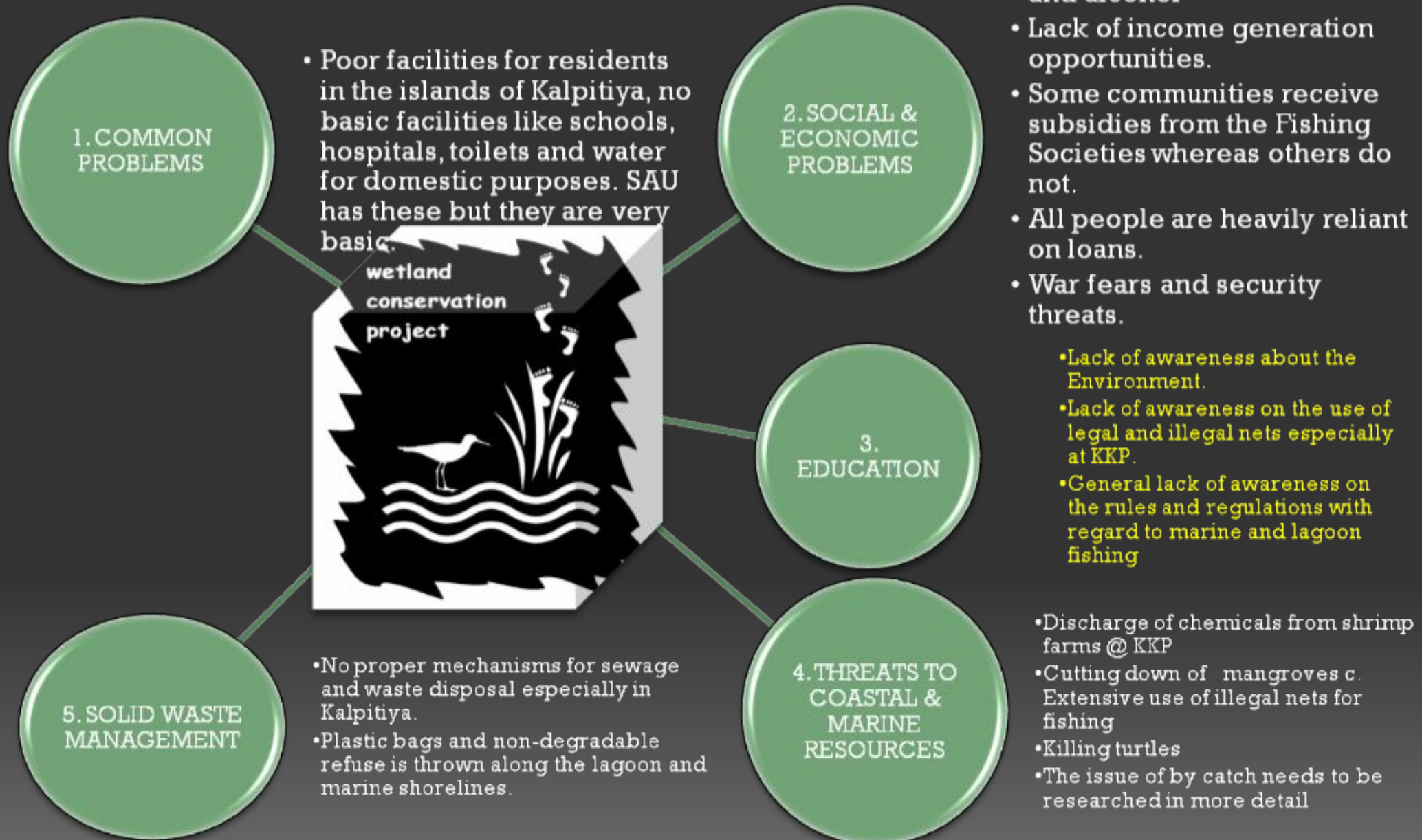
- The project collected primary data on

- Religion
- Ethnicity
- Education
- Access to services (health, electricity, water, sanitation and housing)
- Livelihood systems
- Income fishing practices
- By-catch
- awareness of environmental legislation and fines
- Land tenure
- Gender issues and roles
- Tourism
- The work of Government agencies, CBOs, NGOs and INGOs in the area



Human Impact @ KKP and SAU

- Highlight the ecological, biological and socio-economical importance of KKP to local, national, regional and international conservation efforts -



Methodology and Materials Used

SLWCS Comprehensive Rapid Evaluation System (CRES) which is an objective research based priority ranking system that utilizes multiple techniques such as

- The DELPHI Technique
- Participatory Rural Appraisals
- Rapid Rural Appraisals
- Central Environmental Authority (SL-CEA) Wetland Evaluation System
- North American Wetlands Conservation Councils –Canada Method
- RECCE Method for New Zealand Surveys
- Broad Reconnaissance Survey
- Point Count Method

At national and provincial levels taking the village (Grama Niladhari Division) and community members as the smallest unit of information.

CRES ensures that a site is evaluated based on its natural, socio-economic and cultural values in a way that sites and sub-sites can be compared and actions prioritized.

It has been utilized in 7 sites in 4 administrative provinces of Sri Lanka and in Bengkulu Province, Sumatra, Indonesia.

Discussion

Attributes

KKP

SAU 4.25

Biological Diversity

4.8: This is mainly due to the presence of threatened animal and plant species

4: Due to unique Habitat and high bird diversity near water bodies.

Scenic & Landscape

4.8: Due to unique Habitat and Ecology –threatened animals and plant species along with critical habitats for migratory birds.

4: Due to landscape beauty of the wetland stands at a high level with uniquely diverse and important habitats.

Historical & Cultural

4: Due to historical significance, recreational values and educational and scientific importance.

5: Due to high archeological and cultural significance and is being proposed as a UNESCO World Heritage Site.

Uniqueness

4.8: Due to unique Habitat and Ecology –threatened animals and plant species along with critical habitats for migratory birds.

4: Due to high percentage of avifauna and all the major faunal groups are represented.

Urgent Actions

Initiate environmental education programs such as the Outdoor Environmental Education Program and the Mobile Environmental Education Workshop to educate community members and school children about biodiversity conservation and the importance of sustainably managing the natural environment.

Highlight the importance of giving coastal and marine ecosystems the same considerations given to freshwater systems

Disseminate of the report and organize forums exhibitions, workshops and presentations to educate the general public, bureaucrats, and policy makers. CEA-IWMI Wetland Database for the use of scientists and planners.

Conduct meetings with stakeholders

- the regional and national level
- the Delphi technique will be utilized to establish relative importance factors

Strengthen and support the Department of Wildlife Conservation and the Forest Department to actively protect the declared sites while allowing sustainable resource utilization.

Urgent Action

Develop indicators to measure wetlands in Sri Lanka,

- using a combination of methods that were used for this project.

Recommend to the Wetlands Commission and other government institutions to have the KKP and SAU Coastal Wetland Complexes listed as a Ramsar wetlands and Important Bird Areas

Establish a long term monitoring program.

Apply the Hydrogeomorphic Approach and the Integrated Bio-Economic Planning System of wetland evaluation.

- These are science-based tools for measuring wetlands and their bio-economic health. These systems use spatial, environmental and economic analysis and require exhaustive data collection and analysis.

Establish a historic and current land-use change analysis program using GIS and satellite imagery

Develop management strategies for the KKP to promote sustainable use of resources that will ensure livelihood security as well as financial stability. These efforts must provide incentives as a way of encouraging communities to give up harmful and unsustainable resource extraction.

Acknowledgements

- Ocean Park Conservation Fund, IUCN Netherlands Committee, UNESCO-SL and Escape Tourism for their financial support of the project.
- Mr. Tilak Conrad, Kalpitiya Naval Base Commander, the Sri Lanka Police and the three Divisional Secretaries of Puttalam, Kalpitiya and Karaitive.
- Mr. Anurudha Seneviratne, Secretary of the World Heritage Preservation Committee. 225 Brigade of the 22nd Division of the Sri Lanka Army and the Venerable Seruwila Saranakeerthi Thera.
- Additionally we would like to thank our logistics staff, international volunteers, local helpers, boatmen, Grama Niladari Officer, villagers, and all the other numerous individuals from various organizations who assisted us to conduct this study.

Acknowledgements

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The full and comprehensive report (along with similar reports for Bellanwilla/Attidiya, Flood Plains and Bandaragama Wetlands) are published and available at www.slwcs.org/research as part of our “Wetland Conservation Project (WCP)”

Hard copies of SLWCS reports, articles and scientific/nature publications can be referred to and obtained from the library at the Society’s head office

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